Directions: Write answers on this page and show any work on separate pages.

Problem 1: Compute the following indefinite integrals. In each case, assume n is constant. Show work by applying integration techniques. $\left(Recall : i = \sqrt{-1} \implies i \text{ is a constant}\right)$

(a) $\displaystyle\int \sin(nx)\,dx =$

(b) $\displaystyle\int e^{inx}\,dx =$

(c) $\displaystyle\int xe^{inx}\,dx =$

(d) $\displaystyle\int x^2e^{inx}\,dx =$

Problem 2: Determine all possible values of the expressions below for $n = 1, 2, 3, ...$ by rewriting each expression as a piece-wise function of n or preferably as an equivalent single function of n. (*Hint: Try some values of n to see a pattern*)

(a) $\sin(\pi n) =$ $\quad$ $\cos(\pi n) =$ $\quad$ $e^{i\pi n} =$

(b) $\sin(2\pi n) =$ $\quad$ $\cos(2\pi n) =$ $\quad$ $e^{i2\pi n} =$

(c) $\sin\left(\frac{\pi}{2}n\right) =$ $\quad$ $\cos\left(\frac{\pi}{2}n\right) =$ $\quad$ $e^{i\pi n/2} =$

Problem 3: Circle the correct answer in brackets [ ] and/or fill in any blanks.

(a) If $f(x)$ is an even function then ________________ (*Look up the definition somewhere*)

(b) If $f(x)$ is an odd function then ________________

(c) If $f(x)$ is an even function then $\displaystyle\int_{-L}^{L} f(x)\,dx =$

(d) If $f(x)$ is an odd function then $\displaystyle\int_{-L}^{L} f(x)\,dx =$

(e) Cosine is an [EVEN / ODD] function, therefore $\cos(-x) =$

(f) Sine is an [EVEN / ODD] function, therefore $\sin(-x) =$

(g) By Euler's formula and (e) & (f), $e^{-inx} + e^{inx} =$

(h) By Euler's formula and (e) & (f), $(e^{-inx} - e^{inx})i =$